

Fifth Section.
The Modular Functions.

§ 51. The elliptic differential equations.

The elliptic functions defined up to this point are functions of the two variables v, ω , and the moduli κ, κ' , furthermore $j(\omega), K, K'$, depend on the period-ratio ω , which must have a positive imaginary part. Likewise the function $\wp(u)$ is a function of the three variables u, ω_1, ω_2 , and the invariants g_2, g_3 depend on ω_1, ω_2 . But the problem of the inversion of the elliptic integrals, as we formulated it in § 14, presupposes that in the elliptic functions κ^2 (or, in the Weierstrass normal form, g_2, g_3) have arbitrarily given values; the complete solution of this problem therefore requires that not ω , but κ^2 (respectively g_2, g_3), appear as the second independent variable, and that ω be expressed by means of these. The next considerations will be devoted to this problem.

We start from the problem of integrating a system of differential equations, which we call the elliptic differential equations:

$$\frac{dx}{dv} = yz, \quad \frac{dy}{dv} = -zx, \quad \frac{dz}{dv} = -\kappa^2 xy, \quad (1)$$

under the subsidiary conditions:

$$\text{for } v = 0 \text{ one shall have } x = 0, \quad y = 1, \quad z = 1. \quad (2)$$

We have seen in the preceding section that, when

$$\kappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \quad (3)$$

this problem is solved by the elliptic functions, and indeed in such a way that x, y, z become single-valued and, except where they are infinite, continuous functions of v in the whole v -plane. But if κ^2 is now given, the question arises whether for every value of κ^2 there is a value of ω satisfying the condition (3), and whether there are several such values.

We first prove that the functions x, y, z are uniquely determined by the differential equations (1) together with the subsidiary conditions.

It follows first from (1) that $x^2 + y^2$ and $\kappa^2 x^2 + z^2$ are constant, and from the subsidiary conditions one obtains

$$y^2 = 1 - x^2, \quad z^2 = 1 - \kappa^2 x^2. \quad (4)$$

Suppose there exists a second system of functions x_1, y_1, z_1 satisfying the same conditions. Then first

$$y_1^2 = 1 - x_1^2, \quad z_1^2 = 1 - \kappa^2 x_1^2,$$

and a simple differentiation gives that

$$\frac{xy_1z_1 - x_1yz}{1 - \kappa^2 x^2 x_1^2}$$

is constant. (The addition theorem of the elliptic integrals, from which one obtains the form of this expression, is confirmed immediately by differentiation.) From the subsidiary conditions, however, it follows that this constant vanishes, hence

$$xy_1z_1 = x_1yz,$$

from which one concludes easily, by squaring both sides,

$$x = x_1, \quad y = y_1, \quad z = z_1.$$

§ 52. The independent variable $\varkappa^2$. Linear differential equation for K .

We now first show that, and how, for an arbitrarily given value of $\varkappa^2$, at least one value of ω satisfying the condition

$$\varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4} \quad (1)$$

can be found. This question is answered most completely by reduction to a linear differential equation of the second order.

Such a differential equation is already contained in the formulae of § 23. It follows first from § 23, (14):

$$d \log \varkappa^2 = \pi i \vartheta_{01}^4 d\omega,$$

or, with regard to

$$\pi \vartheta_{00}^2 = 2K, \quad \pi \vartheta_{10}^2 = 2\varkappa K, \quad \pi \vartheta_{01}^2 = 2\varkappa' K \quad [\S 42, (3), (15)], \quad (2)$$

$$\pi d(\varkappa^2) = 4i\varkappa^2 \varkappa'^2 K^2 d\omega, \quad (3)$$

and from § 23, (19):

$$\frac{d}{d\omega} \frac{1}{K^2} \frac{dK}{d\omega} = -\frac{4}{\pi^2} \varkappa^2 \varkappa'^2 K^3.$$

If, by means of (3), one introduces the variable $\varkappa^2$ in place of ω , there follows

$$\frac{d}{d(\varkappa^2)} \left(\varkappa^2 \varkappa'^2 \frac{dK}{d(\varkappa^2)} \right) = \frac{1}{4} K, \quad (4)$$

and this is the desired differential equation.

Put further

$$-i\omega = \frac{K'}{K}, \quad (5)$$

$$-iK^2 d\omega = K dK' - K' dK. \quad (6)$$

Then by (3) one obtains

$$\varkappa^2 \varkappa'^2 \left(K \frac{dK'}{d(\varkappa^2)} - K' \frac{dK}{d(\varkappa^2)} \right) = -\frac{\pi}{4}, \quad (7)$$

and therefore, by differentiating once more,

$$\frac{1}{K} \frac{d}{d(\varkappa^2)} \left(\varkappa^2 \varkappa'^2 \frac{dK}{d(\varkappa^2)} \right) = \frac{1}{K'} \frac{d}{d(\varkappa^2)} \left(\varkappa^2 \varkappa'^2 \frac{dK'}{d(\varkappa^2)} \right),$$

from which one sees that K' is the second particular integral of the differential equation (4).

This differential equation can be integrated by hypergeometric series (Gauss, *Disq. circa seriem infinitum*, Werke, Vol. III, p. 125).

If for the moment we put $\varkappa^2 = x$, $K = y$, then the differential equation (4) reads

$$x(1-x) \frac{d^2 y}{dx^2} + (1-2x) \frac{dy}{dx} - \frac{1}{4} y = 0, \quad (8)$$

and it arises from the general Gaussian differential equation

$$(x-x^2) \frac{d^2 y}{dx^2} + [\gamma - (\alpha + \beta + 1)x] \frac{dy}{dx} - \alpha\beta y = 0 \quad (9)$$

when one sets $\alpha = \beta = \frac{1}{2}$, $\gamma = 1$. Hence one of its particular integrals is

$$\begin{aligned} F\left(\frac{1}{2}, \frac{1}{2}, 1, x\right) &= 1 + \sum_{\nu=1}^{\infty} \left(\frac{1 \cdot 3 \cdot 5 \cdots (2\nu-1)}{2 \cdot 4 \cdots 2\nu} \right)^2 x^{\nu} \\ &= \sum_{\nu=0}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(\frac{x}{16} \right)^{\nu}, \end{aligned} \quad (10)$$

and this series is convergent so long as the absolute value of x is smaller than 1. As the second particular integral one may choose

$$F\left(\frac{1}{2}, \frac{1}{2}, 1, 1-x\right), \quad (11)$$

but it has another convergence domain. If x is represented as a complex variable in a plane, then (10) converges in a circle described around the origin with radius 1, and (11) in an equal circle described around the point 1, so that the common convergence domain consists of a lune containing the part of the real axis lying between 0 and 1. But one can also, which is important for us, set up the second particular integral in a form which converges in the same circle as the series (10). In doing this, one should observe that the second particular integral becomes infinite for $x = 0$.

This second particular integral is obtained by the following passage to the limit. If we denote the hypergeometric series according to Gauss by

$$\begin{aligned} F(\alpha, \beta, \gamma, x) &= 1 + \frac{\alpha\beta}{1 \cdot \gamma} x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1 \cdot 2 \cdot \gamma(\gamma+1)} x^2 + \cdots \\ &= \frac{\Pi(\gamma-1)}{\Pi(\alpha-1)\Pi(\beta-1)} \sum_{\nu=0}^{\infty} \frac{\Pi(\alpha+\nu-1)\Pi(\beta+\nu-1)}{\Pi(\gamma+\nu-1)\Pi(\nu)} x^{\nu}, \end{aligned}$$

then in general one obtains the two particular integrals of (9):

$$y_1 = F_1(x) = F(\alpha, \beta, \gamma, x),$$

$$y_2 = F_2'(x) = x^{1-\gamma}(1-x)^{\gamma-\alpha-\beta} F(1-\alpha, 1-\beta, 2-\gamma, x).$$

Since these, however, become identical for $\alpha = \beta = \frac{1}{2}$, $\gamma = 1$, one first puts $\alpha = \beta = \frac{1}{2}$, $\gamma = 1 + \varepsilon$, hence

$$y_1 = F\left(\frac{1}{2}, \frac{1}{2}, 1 + \varepsilon, x\right), \quad y_2 = \left(\frac{1-x}{x}\right)^{\varepsilon} F\left(\frac{1}{2}, \frac{1}{2}, 1 - \varepsilon, x\right).$$

But

$$F\left(\frac{1}{2}, \frac{1}{2}, 1 \pm \varepsilon, x\right) = \frac{\Pi(\pm\varepsilon)}{\Pi\left(-\frac{1}{2}\right)^2} \sum_{\nu=0}^{\infty} \frac{\Pi\left(\nu - \frac{1}{2}\right)^2}{\Pi(\nu \pm \varepsilon)\Pi(\nu)} x^{\nu},$$

and, adding a constant factor, one may take the following also as the second particular integral:

$$\frac{1}{2\varepsilon} \sum_{\nu=0}^{\infty} \frac{\Pi\left(\nu - \frac{1}{2}\right)^2}{\Pi(\nu)} \left[-\frac{1}{\Pi(\nu + \varepsilon)} + \left(\frac{x}{1-x}\right)^{-\varepsilon} \frac{1}{\Pi(\nu - \varepsilon)} \right] x^{\nu}.$$

If this is developed according to powers of ε and then $\varepsilon = 0$ is put, it follows that

$$\sum_{\nu=0}^{\infty} \frac{\Pi\left(\nu - \frac{1}{2}\right)^2}{\Pi(\nu)^2} x^{\nu} \left[\frac{\Pi'(\nu)}{\Pi(\nu)} - \frac{1}{2} \log \frac{x}{1-x} \right],$$

for which, by use of Gauss's formula

$$\frac{\Pi\left(\nu - \frac{1}{2}\right)\Pi(\nu)}{\Pi(2\nu)} = \sqrt{\pi} 2^{-2\nu},$$

one may, omitting a constant factor, also put

$$\sum_{\nu=0}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(\frac{x}{16}\right)^{\nu} \left[\frac{\Pi'(\nu)}{\Pi(\nu)} - \frac{1}{2} \log \frac{x}{1-x} \right]. \quad (12)$$

But for an integral positive ν one has

$$\frac{\Pi'(\nu)}{\Pi(\nu)} = \Pi'(0) + 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{\nu},$$

and if, therefore, for abbreviation we put

$$\begin{aligned} F(x) &= \sum_{\nu=0}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(\frac{x}{16}\right)^{\nu}, \\ G(x) &= 2 \sum_{\nu=1}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{\nu}\right) \left(\frac{x}{16}\right)^{\nu}, \end{aligned} \quad (13)$$

then the two particular solutions of the differential equation (8) are obtained if for the second one takes a linear combination of (10) and (12):

$$y_1 = F(x), \quad y_2 = \frac{1}{2}G(x) - \frac{1}{2} \log \frac{x}{16(1-x)} F(x). \quad (14)$$

But in order to represent the particular integrals K and K' by these functions, we must examine the behavior of K, K' for $q = 0$, for which at the same time $\varkappa^2 = 0$.

For $q = 0$, however, by § 25 and § 42 one has

$$q^{-1} \varkappa^2 = 16, \quad 2K = \pi,$$

hence

$$\lim \left(\log \frac{\varkappa^2}{16} - i\pi\omega \right) = 0;$$

on the other hand, $2iK' = \pi \vartheta_{00}^2 \omega$ (§ 42), or

$$2K' + \log \frac{\varkappa^2}{16} = \left(\log \frac{\varkappa^2}{16} - i\pi\omega \right) + i\pi\omega(1 - \vartheta_{00}^2),$$

from which

$$\lim_{\varkappa^2=0} \left(K' + \frac{1}{2} \log \frac{\varkappa^2}{16} \right) = 0. \quad (15)$$

Since now K and K' are particular integrals of (8) (for $x = \varkappa^2$), they both have the form

$$K = a_1 y_1 + a_2 y_2, \quad K' = a'_1 y_1 + a'_2 y_2,$$

where a_1, a_2, a'_1, a'_2 are constant. Since K remains finite for $x = 0$ and assumes the value $\frac{1}{2}\pi$, one has $a_2 = 0$, $a_1 = \frac{1}{2}\pi$, and in order that $K' + \frac{1}{2} \log x$ remain finite and, according to (15), assume the value $\log 4$, one must have $a'_2 = 1$ and $a'_1 = 0$. Thus one obtains, if $x = \varkappa^2$, $1 - x = \varkappa'^2$ is put:

$$K = \frac{\pi}{2} F(\varkappa^2), \quad K' = \frac{1}{2} G(\varkappa^2) - \frac{1}{2} \log \frac{\varkappa^2}{16\varkappa'^2} F(\varkappa^2), \quad (16)$$

and

$$q = e^{-\pi K'/K} = \frac{\varkappa^2}{16\varkappa'^2} e^{-\frac{G(\varkappa^2)}{F(\varkappa^2)}}. \quad (17)$$

Here, as already remarked, the series $G(\varkappa^2)$, $F(\varkappa^2)$ are convergent so long as the absolute value of $\varkappa^2$ is a proper fraction. In the differential equation (4) there are, to be sure, the means of continuing

the expressions found for K, K', q beyond this convergence domain. But one arrives more simply by applying Landen's transformation.

We bound the plane of the complex values $\varkappa^2$ by making two cuts along the real axis, from 0 to $-\infty$ and from 1 to ∞ . In the plane thus bounded all roots out of $\varkappa^2, \varkappa'^2$ are then uniquely determined by the condition that for positive proper fractional $\varkappa^2$ they shall be real and positive. Now, by the formulae of Landen's transformation [§ 44, (28)], if with $\varkappa_1, \varkappa'_1, K_1, K'_1$ we denote the functions of ω which arise from $\varkappa, \varkappa', K, K'$ when ω is replaced by 2ω , then

$$\varkappa_1 = \frac{1 - \varkappa'}{1 + \varkappa'}, \quad \varkappa'_1 = \frac{2\sqrt{\varkappa'}}{1 + \varkappa'}, \quad 2K_1 = (1 + \varkappa')K, \quad K'_1 = (1 + \varkappa')K'. \quad (18)$$

Therefore, if in (17) one replaces ω by 2ω and takes the square root, there follows

$$q = \frac{1}{4} \frac{\varkappa_1}{\varkappa'_1} e^{-\frac{1}{2} \frac{G(\varkappa_1^2)}{F(\varkappa_1^2)}}. \quad (19)$$

Here the convergence domain now extends over the unit circle situated in the plane $\varkappa_1^2$, and this corresponds to the whole plane $\varkappa^2$. For the real part of $\varkappa'$ is positive in the whole plane of $\varkappa^2$, except on the two banks of the cut running from $+1$ to $+\infty$, on which $\varkappa'$ is purely imaginary. From this it follows that the absolute value of $\varkappa_1$ is smaller than 1, and that the circumference of the unit circle in the plane $\varkappa_1$ corresponds to the two banks of this cut.

But one can still improve the convergence of these expressions by applying Landen's transformation once more. If we denote the result of a second doubling of ω by the index 2, there follows

$$\sqrt{\varkappa_2} = \frac{1 - \sqrt{\varkappa'}}{1 + \sqrt{\varkappa'}}, \quad 4K_2 = (1 + \sqrt{\varkappa'})^2 K, \quad K'_2 = (1 + \sqrt{\varkappa'})^2 K', \quad q = \sqrt{\frac{\varkappa_2}{4\varkappa'_2}} e^{-\frac{1}{4} \frac{G(\varkappa_2^2)}{F(\varkappa_2^2)}}, \quad (20)$$

where now the convergence domain covers the plane of $\varkappa^2$ twice (with a ramification at the point $\varkappa^2 = 1$).

This operation can be continued, by deriving from $\varkappa_2$, by doubling ω , a new quantity $\varkappa_3$, from this likewise $\varkappa_4$, and so forth, according to the formula (17). One then obtains a series of expressions for q , whose convergence becomes better and better. For an arbitrary ν one has

$$q = \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4\varkappa'_\nu}} e^{-\frac{1}{2^\nu} \frac{G(\varkappa_\nu^2)}{F(\varkappa_\nu^2)}}. \quad (21)$$

The first equation (18),

$$\varkappa_1 = \frac{\varkappa^2}{(1 + \varkappa')^2},$$

shows that so long as the absolute value of $\varkappa^2$ is smaller than 1, and consequently the absolute value of $1 + \varkappa'$ is greater than 1, the absolute value of $\varkappa_1$ is smaller than that of $\varkappa^2$, and consequently $\varkappa_\nu$ approaches the limit zero as ν increases without bound. From this one obtains for q the representation

$$q = \lim_{\nu=\infty} \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4\varkappa'_\nu}} = \lim_{\nu=\infty} \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4}}, \quad (22)$$

which is suitable for calculation at real values of $\varkappa$.

From each of the formulae (21) one can derive an expansion of q according to the ascending powers of $\varkappa_\nu$, whose convergence increases with increasing ν . If in particular one applies formula (20), one obtains an expansion communicated by Weierstrass in the monthly reports of the Berlin Academy for the year 1883, whose convergence domain fills the plane $\varkappa^2$ twice.

The coefficients of this expansion are most simply computed from

$$\sqrt{\varkappa_2} = \frac{\vartheta_{10}(0, 4\omega)}{\vartheta_{00}(0, 4\omega)} = \frac{2q + 2q^9 + 2q^{25} + \dots}{1 + 2q^4 + 2q^{16} + \dots}, \quad (23)$$

and in this way one obtains, by the method of undetermined coefficients,

$$q = \frac{\sqrt{\varkappa_2}}{2} + 2 \left(\frac{\sqrt{\varkappa_2}}{2} \right)^5 + 15 \left(\frac{\sqrt{\varkappa_2}}{2} \right)^9 + 150 \left(\frac{\sqrt{\varkappa_2}}{2} \right)^{13} + 1707 \left(\frac{\sqrt{\varkappa_2}}{2} \right)^{17} + \dots. \quad (24)$$

Likewise, from (23), conversely, $\sqrt{\varkappa_2}$ can be developed in a series proceeding according to powers of q :

$$\frac{1}{2}\sqrt{\varkappa_2} = q - 2q^5 + 5q^9 - 10q^{13} + 18q^{17} \dots, \quad (25)$$

which one also obtains by inversion of the series (24).

With this the question posed at the beginning of this paragraph is completely answered.

With regard to the application of the formulae developed here to the numerical calculation of q , it should also be noted that, when $\varkappa^2$ lies nearer to 1, one obtains a better-converging procedure by interchanging $\varkappa^2$ with $\varkappa'^2$. The calculation then first gives not q , but

$$q' = e^{-\pi i/\omega},$$

from which, however, one obtains q from the formula

$$\log q \log q' = \pi^2.$$

§ 53. The solutions of the equation $j(\omega) = j(\omega')$.

The results of § 51 are sufficient first to determine the relations of the different values of ω which lead to the same value of $\varkappa^2$. Thus let

$$\frac{\vartheta_{10}^4(0, \omega')}{\vartheta_{00}^4(0, \omega')} = \frac{\vartheta_{10}^4(0, \omega)}{\vartheta_{00}^4(0, \omega)} = \varkappa^2. \quad (1)$$

Put

$$v = \pi \vartheta_{00}^2(0, \omega) u = \pi \vartheta_{00}^2(0, \omega') u'. \quad (2)$$

Then (by § 42) both the functions

$$\frac{\vartheta_{00}(0, \omega) \vartheta_{11}(u, \omega)}{\vartheta_{10}(0, \omega) \vartheta_{01}(u, \omega)}, \quad \frac{\vartheta_{01}(0, \omega) \vartheta_{10}(u, \omega)}{\vartheta_{10}(0, \omega) \vartheta_{01}(u, \omega)}, \quad \frac{\vartheta_{01}(0, \omega) \vartheta_{00}(u, \omega)}{\vartheta_{00}(0, \omega) \vartheta_{01}(u, \omega)}, \quad (3)$$

and also

$$\frac{\vartheta_{00}(0, \omega') \vartheta_{11}(u', \omega')}{\vartheta_{10}(0, \omega') \vartheta_{01}(u', \omega')}, \quad \frac{\vartheta_{01}(0, \omega') \vartheta_{10}(u', \omega')}{\vartheta_{10}(0, \omega') \vartheta_{01}(u', \omega')}, \quad \frac{\vartheta_{01}(0, \omega') \vartheta_{00}(u', \omega')}{\vartheta_{00}(0, \omega') \vartheta_{01}(u', \omega')}, \quad (4)$$

when put for x, y, z , satisfy the differential equations (1) of the preceding paragraph, and the corresponding functions among these are therefore identical. If now $u' = \frac{1}{2}$, then $\vartheta_{10}(u', \omega')$ vanishes, and therefore $\vartheta_{10}(u, \omega)$ must also vanish. Hence by (2), with regard to § 21,

$$\vartheta_{00}^2(0, \omega') = \vartheta_{00}^2(0, \omega)(\alpha + \beta\omega), \quad (5)$$

where α, β are whole numbers which satisfy the condition

$$\alpha \equiv 1, \quad \beta \equiv 0 \pmod{2}. \quad (6)$$

If $u' = \omega'/2$, then $\vartheta_{01}(u', \omega')$ and $\vartheta_{01}(u, \omega)$ vanish, and therefore, as above,

$$\vartheta_{00}^2(0, \omega') \omega' = \vartheta_{00}^2(0, \omega)(\gamma + \delta\omega), \quad (7)$$

where

$$\gamma \equiv 0, \quad \delta \equiv 1 \pmod{2}. \quad (8)$$

From this follows

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \quad (9)$$

But since in the same way one can conclude

$$\vartheta_{00}^2(0, \omega) = \vartheta_{00}^2(0, \omega')(\alpha' + \beta'\omega'),$$

$$\vartheta_{00}^2(0, \omega)\omega = \vartheta_{00}^2(0, \omega')(\gamma' + \delta'\omega'),$$

where $\alpha', \beta', \gamma', \delta'$ are likewise whole numbers, substitution of (5) in these equations gives

$$1 = (\alpha + \beta\omega)(\alpha' + \beta'\omega'), \quad \omega = (\alpha + \beta\omega)(\gamma' + \delta'\omega').$$

If in these equations ω' is expressed by ω according to (9), one obtains

$$1 = \alpha'(\alpha + \beta\omega) + \beta'(\gamma + \delta\omega),$$

$$\omega = \gamma'(\alpha + \beta\omega) + \delta'(\gamma + \delta\omega),$$

and since ω is not real, each of these equations splits into two others:

$$\alpha\alpha' + \gamma\beta' = 1, \quad \beta\alpha' + \delta\beta' = 0,$$

$$\alpha\gamma' + \gamma\delta' = 0, \quad \beta\gamma' + \delta\delta' = 1.$$

Hence

$$(\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma') = 1,$$

and therefore, since ω, ω' must both have a positive imaginary part, and hence by (9) the determinant $\alpha\delta - \beta\gamma$ must be positive,

$$\alpha\delta - \beta\gamma = 1, \quad \alpha = \delta', \quad \delta = \alpha', \quad \gamma = -\gamma', \quad \beta = -\beta'.$$

1. Thus ω, ω' are connected with one another by a linear transformation, and indeed, by (6), (8), by one of the first class (§ 36). Conversely, it has already been proved above that two such values ω, ω' give the same value $\varkappa^2$.

A similar theorem results as an immediate consequence of this for the invariant $j(\omega)$ (§ 46).

2. The equation

$$j(\omega') = j(\omega) \quad (10)$$

is satisfied if and only if ω' is connected with ω by some linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

that is, if

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}, \quad \alpha\delta - \beta\gamma = 1. \quad (11)$$

For if we denote by λ^2, λ'^2 the values of $\varkappa^2, \varkappa'^2$ belonging to ω', ω , then equation (10) can be written

$$\frac{(1 - \lambda^2\lambda'^2)^3}{\lambda^4\lambda'^4} = \frac{(1 - \varkappa^2\varkappa'^2)^3}{\varkappa^4\varkappa'^4},$$

and with respect to λ^2 it is an equation of the sixth degree, whose roots are

$$\varkappa^2, \quad \varkappa'^2, \quad \frac{1}{\varkappa^2}, \quad \frac{1}{\varkappa'^2}, \quad -\frac{\varkappa'^2}{\varkappa^2}, \quad -\frac{\varkappa^2}{\varkappa'^2}.$$

With regard to § 45, therefore, one of the following relations holds:

$$\lambda^2 = \varkappa^2(\omega), \quad \varkappa^2\left(-\frac{1}{\omega}\right), \quad \varkappa^2(\omega+1), \quad \varkappa^2\left(\frac{\omega-1}{\omega}\right), \quad \varkappa^2\left(-\frac{1}{\omega+1}\right), \quad \varkappa^2\left(\frac{\omega}{1-\omega}\right),$$

from which the correctness of the second theorem follows from the first.

The variable ω is here always restricted to a positive imaginary part, and if we call two numbers ω, ω' connected by an equation (11) equivalent, then all numbers equivalent to ω have a positive imaginary part. We can also express our theorem 2 as follows:

3. The function $j(\omega)$ has one and the same value for all equivalent values ω , and only for these.¹

To every value of ω belongs a definite value of $j(\omega)$, and to every value of $j(\omega)$ a value of ω and the totality of equivalent values ω' . Therefore if a single-valued function $\Phi(\omega)$ of ω satisfies the condition

$$\Phi\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \Phi(\omega),$$

then $\Phi(\omega)$ is a single-valued function of $j(\omega)$; and if we therefore assume that $\Phi(\omega)$ becomes infinite for only a finite number of values $j(\omega)$, and becomes infinite only to finite order, it follows that $\Phi(\omega)$ is a rational function of $j(\omega)$.

The supposition is fulfilled, for example, whenever $\Phi(\omega)$ stands in an algebraic connection with $j(\omega)$.

§ 54. The modular functions.

Let $\psi(\omega)$ be a single-valued function of ω . If one applies to ω a linear transformation

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

then $\psi(\omega)$ goes over into another function

$$\psi\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right), \tag{1}$$

which we shall denote by $\psi | S$.

The function $\psi(\omega)$ may possibly remain unchanged under certain transformations S (for instance, always under the identical transformation). All transformations which leave a function $\psi(\omega)$ unchanged form a group $\mathfrak{G}$. For if

$$\psi | S = \psi, \quad \psi | S_1 = \psi,$$

then also

$$\psi | SS_1 = \psi | S_1 = \psi.$$

The group $\mathfrak{G}$ is a divisor of the group $\mathfrak{L}$ of all linear transformations.

If $\mathfrak{G}$ is a divisor of $\mathfrak{L}$ of finite index $(\mathfrak{L}, \mathfrak{G})$, and $\psi(\omega)$ is a function of ω with the property that conversely

$$\psi(\omega') = \psi(\omega)$$

implies a linear connection

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

¹This theorem was first proved by Dedekind (Crelle, Vol. 83). Accordingly Dedekind calls the function $\text{val}(\omega) = 3^{-3}2^{-6}j(\omega)$ the valence of ω .

in which

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

is a transformation belonging to $\mathfrak{G}$, then $\psi(\omega)$ is called a modular function belonging to the group $\mathfrak{G}$.

We restrict ourselves to the consideration of such modular functions as stand in an algebraic connection with the modulus $\varkappa^2$, and hence also with $j(\omega)$.

Under this supposition the following general theorem can be stated:

If $\psi(\omega)$ belongs to the group $\mathfrak{G}$, and $\chi(\omega)$ remains unchanged under the transformations of $\mathfrak{G}$, then $\chi(\omega)$ can be expressed rationally through $\psi(\omega)$.

For first $\chi(\omega)$ is an algebraic function of $\psi(\omega)$, and $\psi(\omega)$, as an algebraic function of $j(\omega)$, can assume every value. But by supposition only one value of $\chi(\omega)$ belongs to every value of $\psi(\omega)$, and therefore $\chi(\omega)$, as a single-valued algebraic function of $\psi(\omega)$, is rational.

The study of the groups contained in $\mathfrak{L}$ and of the modular functions belonging to them forms the main subject of the great work of Klein and Fricke, *Vorlesungen über die Theorie der elliptischen Modulfunktionen* (2 vols., Leipzig, Teubner, 1890, 1892). Here we consider only some quite special cases of these groups and functions which we have encountered in the course of our investigations.

A large class of divisors of the group $\mathfrak{L}$ with finite index is formed by the so-called congruence groups, which, if m is any whole number, are characterized by the four congruences

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{m}.$$

The modular functions belonging to these groups are called congruence moduli of the m -th stage. For $m = 2$ this is the group $\mathfrak{A}$ (§ 36). We shall make use of the following theorems:

1. The modulus $\varkappa^2$ is, by § 52, a modular function belonging to this group; since, by § 36, all these transformations can be composed from the two

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

we can state the theorem:

Every modular function which remains unchanged under the two interchanges

$$(\omega, \omega + 2), \quad \left(\omega, \frac{\omega}{1 + 2\omega}\right),$$

is a rational function of $\varkappa^2$.

2. The function $\varkappa^2 \varkappa'^2$ belongs to the group $\mathfrak{A}'$ formed in common from the first and second classes of § 36. This group can be derived by repetition of the two transformations

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence we can state the theorem:

Every modular function which remains unchanged under the two interchanges

$$(\omega, \omega + 2), \quad \left(\omega, -\frac{1}{\omega}\right),$$

is a rational function of $\varkappa^2 \varkappa'^2$. From this there further follows:

3. Every modular function which remains unchanged under $(\omega, \omega + 2)$ and changes its sign under $\left(\omega, -\frac{1}{\omega}\right)$ is the product of $(\varkappa'^2 - \varkappa^2)$ with a rational function of $\varkappa^2 \varkappa'^2$.

4. The invariant $j(\omega)$ belongs to the group $\mathfrak{L}$ (§ 53), and hence the theorem:
Every modular function which remains unchanged under the two interchanges

$$(\omega, \omega + 1), \quad \left(\omega, -\frac{1}{\omega}\right)$$

is a rational function of $j(\omega)$.

Besides these we introduce two other modular functions. From the definition of the functions $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ [§ 34,(1),(10)] it follows, if the two equations § 42,(3) are multiplied by one another, that

$$f(\omega) = \frac{\sqrt[16]{2}}{\sqrt[12]{\kappa\kappa'}}, \quad f_1(\omega) = \sqrt[16]{2} \sqrt[12]{\frac{\kappa'^2}{\kappa}}, \quad f_2(\omega) = \sqrt[16]{2} \sqrt[12]{\frac{\kappa^2}{\kappa'}}, \quad (2)$$

$$\frac{f_1(\omega)}{f(\omega)} = \sqrt[4]{\kappa'}, \quad \frac{f_2(\omega)}{f(\omega)} = \sqrt[4]{\kappa}. \quad (3)$$

On the basis of § 46,(15),(16) we define two functions $\gamma_2(\omega)$, $\gamma_3(\omega)$:

$$\begin{aligned} \gamma_2(\omega) &= \sqrt[3]{j(\omega)} = \sqrt[3]{2^8} \frac{1 - \kappa^2 \kappa'^2}{\sqrt[3]{\kappa^4 \kappa'^4}}, \\ \gamma_3(\omega) &= \sqrt{j(\omega) - 27 \cdot 64} = \frac{8(2 + \kappa^2 \kappa'^2)(\kappa'^2 - \kappa^2)}{\kappa^2 \kappa'^2}, \end{aligned} \quad (4)$$

which, by (2) and (3), can be represented as single-valued functions of ω as follows:

$$\gamma_2(\omega) = \frac{f(\omega)^{24} - 16}{f(\omega)^8}, \quad \gamma_3(\omega) = \frac{[f(\omega)^{24} + 8][f_1(\omega)^8 - f_2(\omega)^8]}{f(\omega)^8}, \quad (5)$$

for which, according to the fundamental formulae for the f -functions [§ 34,(11),(12)], one can also put

$$\gamma_2(\omega) = f(\omega)^8 f_1(\omega)^8 + f(\omega)^8 f_2(\omega)^8 - f_1(\omega)^8 f_2(\omega)^8, \quad (6)$$

$$\gamma_3(\omega) = \frac{1}{2}[f(\omega)^8 - f_1(\omega)^8][f(\omega)^8 + f_2(\omega)^8][f_1(\omega)^8 - f_2(\omega)^8]. \quad (7)$$

Thus $x = f^8, -f_1^8, -f_2^8$ are the roots of the cubic equation

$$x^3 - \gamma_2 x - 16 = 0, \quad (8)$$

and $4\gamma_3^2$ is the discriminant of this equation.

Also noteworthy are the differential equations:

$$d\kappa^2 = -d\kappa'^2 = \pi i \vartheta_{00}^4 \kappa^2 \kappa'^2 d\omega \quad [\S 52, (3)], \quad (9)$$

$$d\kappa^2 \kappa'^2 = -\pi i \vartheta_{00}^4 (\kappa^2 - \kappa'^2) \kappa^2 \kappa'^2 d\omega, \quad (10)$$

$$df(\omega)^8 = \frac{\pi i}{3} \vartheta_{00}^4 [f_2(\omega)^8 - f_1(\omega)^8] d\omega, \quad (11)$$

$$d\gamma_2(\omega) = -\frac{2\pi i}{3} \gamma_3(\omega) \eta(\omega)^4 d\omega \quad [\text{by (5) and } \S 34,(10)], \quad (12)$$

$$dj(\omega) = -2\pi i \gamma_2(\omega)^2 \gamma_3(\omega) \eta(\omega)^4 d\omega. \quad (13)$$

Next one obtains for the linear fundamental transformations, from § 34,(13),(14),(15),

$$\gamma_2(\omega + 1) = e^{-\frac{2\pi i}{3}} \gamma_2(\omega), \quad \gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega), \quad (14a)$$

$$\gamma_3(\omega + 1) = -\gamma_3(\omega), \quad \gamma_3\left(-\frac{1}{\omega}\right) = -\gamma_3(\omega), \quad (14b)$$

and from § 40 one finds in general

$$\gamma_2 \left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega} \right) = \varrho \gamma_2(\omega), \quad \gamma_3 \left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega} \right) = (-1)^{\alpha\gamma + \beta\gamma + \beta\delta} \gamma_3(\omega), \quad (15)$$

where ϱ has the meaning given in § 40,(3):

$$\varrho = e^{-\frac{2\pi i}{3}(\alpha\gamma + \beta\delta - \alpha\beta - \alpha^2\beta\delta)}.$$

These are therefore modular functions, and the groups to which they belong are characterized by the two congruences

$$\begin{aligned} -\alpha\beta + \alpha\gamma + \beta\delta - \alpha^2\beta\gamma &\equiv 0 \pmod{3}, \\ \alpha\gamma + \beta\gamma + \beta\delta &\equiv 0 \pmod{2}. \end{aligned}$$

Accordingly we give our theorem 4 the following supplements, which follow from formulae (14).

5. A modular function which changes its sign under the two substitutions

$$(\omega, \omega + 1), \quad \left(\omega, -\frac{1}{\omega} \right),$$

is the product of $\gamma_3(\omega)$ with a rational function of $j(\omega)$.

6. A modular function which remains unchanged under the substitution $\left(\omega, -\frac{1}{\omega} \right)$ and under $(\omega, \omega + 1)$ assumes the factor $e^{\pm 2\pi i/3}$ is the product of $\gamma_2(\omega)^{\pm 1}$ with a rational function of $j(\omega)$.

7. A modular function which changes its sign under the substitutions $\left(\omega, -\frac{1}{\omega} \right)$ and under $(\omega, \omega + 1)$ assumes the factor $-e^{\pm 2\pi i/3}$ is the product of $\gamma_2(\omega)\gamma_3(\omega)^{\pm 1}$ with a rational function of $j(\omega)$.

We shall occupy ourselves more thoroughly with the modular functions in the second part.

§ 55. Representation of the elliptic functions by v and $\varkappa^2$.

If we conceive the inversion problem of the elliptic integrals, as in § 51, as the problem of integrating the elliptic differential equations, then it requires the representation of the three functions $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$ by the two independent variables $v, \varkappa^2$. This problem is solved by § 42, but only implicitly with respect to the variable $\varkappa^2$. One can also find representations in which $\varkappa^2$ occurs explicitly, namely by series which proceed according to powers of v , whose coefficients are rational functions of $\varkappa^2$. The coefficients of these series, to be sure, cannot be represented by a general law, but can only be computed successively. These representations are due principally to Weierstrass.²

The σ -functions, since they are everywhere finite and continuous functions of u , can be developed in unconditionally convergent power series according to u ; and if we therefore develop the σ -functions occurring in § 45,(2) in this way, we obtain a solution of our problem. Accordingly we put

$$\begin{aligned} \sigma(v, 2K, 2iK') &= \sum_{\nu=0}^{\infty} A_{\nu} \frac{v^{2\nu+1}}{(2\nu+1)!}, \\ \sigma_{00}(v, 2K, 2iK') &= \sum_{\nu=0}^{\infty} B_{\nu} \frac{v^{2\nu}}{(2\nu)!}, \\ \sigma_{01}(v, 2K, 2iK') &= \sum_{\nu=0}^{\infty} C_{\nu} \frac{v^{2\nu}}{(2\nu)!}, \\ \sigma_{10}(v, 2K, 2iK') &= \sum_{\nu=0}^{\infty} D_{\nu} \frac{v^{2\nu}}{(2\nu)!}, \end{aligned} \quad (1)$$

²On the Weierstrass theory of elliptic functions one should compare especially the *Formeln und Lehrsätze zum Gebrauch der elliptischen Funktionen*, edited by H. A. Schwarz. On the series expansions one should also compare Königsberger, *Vorlesungen über die Theorie der elliptischen Funktionen*, Lecture 25.

and the $A_\nu, B_\nu, C_\nu, D_\nu$ are the functions to be determined of $\varkappa^2$ or of ω . If we first regard them as functions of ω , then the linear transformations [§ 45, (3)] give

$$\begin{aligned} A_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu A_\nu(\omega), \\ B_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu B_\nu(\omega), \\ C_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu D_\nu(\omega), \\ D_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu C_\nu(\omega), \end{aligned} \tag{2}$$

$$\begin{aligned} \varkappa'^{2\nu} A_\nu(\omega + 1) &= A_\nu(\omega), \\ \varkappa'^{2\nu} B_\nu(\omega + 1) &= C_\nu(\omega), \\ \varkappa'^{2\nu} C_\nu(\omega + 1) &= B_\nu(\omega), \\ \varkappa'^{2\nu} D_\nu(\omega + 1) &= D_\nu(\omega), \end{aligned} \tag{3}$$

$$A_\nu(\omega + 2) = A_\nu(\omega), \quad B_\nu(\omega + 2) = B_\nu(\omega), \quad C_\nu(\omega + 2) = C_\nu(\omega), \quad D_\nu(\omega + 2) = D_\nu(\omega). \tag{4}$$

From this we conclude the characteristic properties of these coefficients as functions of $\varkappa^2$. First it is clear from the meaning of the coefficients that for every value of $\varkappa^2$, with the possible exception of $\varkappa^2 = 0, 1, \infty$, the coefficients $A_\nu(\varkappa^2), B_\nu(\varkappa^2), C_\nu(\varkappa^2), D_\nu(\varkappa^2)$ are finite and continuous functions of $\varkappa^2$.

But even for $\varkappa^2 = 0$ these functions remain finite, and one can find their values easily by letting q , and hence $\varkappa^2$, pass into zero in the representations of § 37. One then obtains from the developments in § 25, with regard to the fact that by § 34 and § 42 for a vanishing q

$$\eta(\omega) = q^{1/12}, \quad 2K = \pi,$$

that

$$\begin{aligned} e^{v^2/6} \sin v &= \sum_{\nu=0}^{\infty} A_\nu(0) \frac{v^{2\nu+1}}{(2\nu+1)!}, \\ e^{v^2/6} &= \sum_{\nu=0}^{\infty} B_\nu(0) \frac{v^{2\nu}}{(2\nu)!}, \\ e^{v^2/6} &= \sum_{\nu=0}^{\infty} C_\nu(0) \frac{v^{2\nu}}{(2\nu)!}, \\ e^{v^2/6} \cos v &= \sum_{\nu=0}^{\infty} D_\nu(0) \frac{v^{2\nu}}{(2\nu)!}. \end{aligned} \tag{5}$$

The formulae (2) now give

$$\begin{aligned} A_\nu(\varkappa'^2) &= (-1)^\nu A_\nu(\varkappa^2), \\ B_\nu(\varkappa'^2) &= (-1)^\nu B_\nu(\varkappa^2), \\ C_\nu(\varkappa'^2) &= (-1)^\nu D_\nu(\varkappa^2), \\ D_\nu(\varkappa'^2) &= (-1)^\nu C_\nu(\varkappa^2), \end{aligned} \tag{6}$$

from which it follows that these functions also remain finite for $\varkappa^2 = 1$, namely

$$\begin{aligned} A_\nu(1) &= (-1)^\nu A_\nu(0), \\ B_\nu(1) &= (-1)^\nu B_\nu(0), \\ C_\nu(1) &= (-1)^\nu D_\nu(0), \\ D_\nu(1) &= (-1)^\nu C_\nu(0). \end{aligned} \tag{7}$$

The system (3) may further be written in the form

$$\begin{aligned}
 \varkappa'^{2\nu} A_\nu \left(-\frac{\varkappa^2}{\varkappa'^2} \right) &= A_\nu(\varkappa^2), \\
 \varkappa'^{2\nu} B_\nu \left(-\frac{\varkappa^2}{\varkappa'^2} \right) &= C_\nu(\varkappa^2), \\
 \varkappa'^{2\nu} C_\nu \left(-\frac{\varkappa^2}{\varkappa'^2} \right) &= B_\nu(\varkappa^2), \\
 \varkappa'^{2\nu} D_\nu \left(-\frac{\varkappa^2}{\varkappa'^2} \right) &= D_\nu(\varkappa^2),
 \end{aligned} \tag{8}$$

whence for an infinite $\varkappa^2$ one obtains

$$\begin{aligned}
 A_\nu(\varkappa^2) &= \varkappa^{2\nu} A_\nu(0), \\
 B_\nu(\varkappa^2) &= \varkappa^{2\nu} D_\nu(0), \\
 C_\nu(\varkappa^2) &= \varkappa^{2\nu} B_\nu(0), \\
 D_\nu(\varkappa^2) &= \varkappa^{2\nu} C_\nu(0),
 \end{aligned} \tag{9}$$

and from this it is concluded that the coefficients $A_\nu, B_\nu, C_\nu, D_\nu$ are integral rational functions of $\varkappa^2$ of degree ν ; from § 54, 2, 3, it further follows that A_ν, B_ν for even ν are expressed rationally through $\varkappa^2 \varkappa'^2$, while for odd ν they are equal to the product of $(\varkappa'^2 - \varkappa^2)$ with a rational function of $\varkappa^2 \varkappa'^2$.

From B_ν one finds C_ν by means of the formula

$$C_\nu(\varkappa^2) = \varkappa'^{2\nu} B_\nu \left(-\frac{\varkappa^2}{\varkappa'^2} \right), \tag{10}$$

and D_ν by application of (6) and (8):

$$D_\nu(\varkappa^2) = (-1)^\nu C_\nu(\varkappa'^2) = (-1)^\nu \varkappa^{2\nu} B_\nu \left(-\frac{\varkappa'^2}{\varkappa^2} \right). \tag{11}$$

Since from (5) one can easily determine the coefficients $A_\nu(0), B_\nu(0), C_\nu(0), D_\nu(0)$, the coefficients $A_\nu, B_\nu, C_\nu, D_\nu$ result from the system of formulae developed here, without further calculation, up to and including $\nu = 3$.

One finds:

$$\begin{aligned}
 A_0 &= 1, & A_1 &= 0, & A_2 &= -\frac{2}{3}(1 - \varkappa^2 \varkappa'^2), \\
 B_0 &= 1, & B_1 &= \frac{1}{3}(\varkappa'^2 - \varkappa^2), & B_2 &= \frac{1}{3}(1 + 2\varkappa^2 \varkappa'^2), \\
 C_0 &= 1, & C_1 &= \frac{1}{3}(1 + \varkappa^2), & C_2 &= \frac{1}{3}(\varkappa'^4 - 2\varkappa^2), \\
 D_0 &= 1, & D_1 &= -\frac{1}{3}(1 + \varkappa'^2), & D_2 &= \frac{1}{3}(\varkappa^4 - 2\varkappa'^2), \\
 A_3 &= -\frac{8}{9}(\varkappa'^2 - \varkappa^2)(2 + \varkappa^2 \varkappa'^2), \\
 B_3 &= \frac{1}{9}(\varkappa'^2 - \varkappa^2)(5 - 2\varkappa^2 \varkappa'^2), \\
 C_3 &= \frac{1}{9}(1 + \varkappa^2)(5\varkappa'^4 + 2\varkappa^2), \\
 D_3 &= -\frac{1}{9}(1 + \varkappa'^2)(5\varkappa^4 + 2\varkappa'^2).
 \end{aligned}$$

Further coefficients can be computed on the same way, although at greater length, by developing the expressions for the σ -functions in § 32 according to powers of q and, as here the lowest power of q was used, using the higher powers. Weierstrass makes use, for the recurrent computation of the coefficients, of certain partial differential equations, similar to that which we shall derive in the next paragraph for the function σ .

§ 56. Power series for the Weierstrass functions $\wp(u), \sigma(u)$.

The function $\sigma(u)$, considered as a function of u , can, as already remarked, be developed into a series according to powers of u convergent in the whole u -plane; and by § 35,(8),(12) this development has the following form:

$$\sigma(u) = u + \alpha u^3 + \alpha_1 u^5 + \alpha_2 u^7 + \alpha_3 u^9 + \dots \quad (1)$$

The function

$$\wp(u) = -\frac{d^2 \log \sigma(u)}{du^2} = \frac{\sigma'(u)^2 - \sigma(u)\sigma''(u)}{\sigma(u)^2} \quad (2)$$

can likewise be developed according to ascending powers of u , but no longer in a series convergent everywhere; rather this series has a convergence circle. It has the form

$$\wp(u) = \frac{1}{u^2} + a + a_1 u^2 + a_2 u^4 + a_3 u^6 + \dots, \quad (3)$$

and one can determine the coefficients $a, a_1, a_2, a_3, \dots$ successively, as functions of g_2, g_3 , from the differential equation [§ 46,(18)]

$$\wp'(u)^2 = 4\wp(u)^3 - g_2\wp(u) - g_3. \quad (4)$$

First one has

$$\begin{aligned} \wp'(u) &= -\frac{2}{u^3} + 2a_1 u + 4a_2 u^3 + 6a_3 u^5 + \dots, \\ \wp'(u)^2 &= \frac{4}{u^6} - \frac{8a_1}{u^2} - 16a_2 + (4a_1^2 - 24a_3)u^2 + \dots, \end{aligned}$$

and since $\wp'(u)^2$ contains no term with u^{-4} , while in $\wp(u)^3$ the term $3au^{-4}$ occurs, a must be zero. After this there results

$$\begin{aligned} \wp(u)^3 &= \frac{1}{u^6} + \frac{3a_1}{u^2} + 3a_2 + 3(a_3 + a_1^2)u^2 + \dots, \\ \wp(u) &= \frac{1}{u^2} + a_1 u^2 + \dots. \end{aligned}$$

And if one inserts this in equation (4), there follows

$$a_1 = \frac{g_2}{20}, \quad a_2 = \frac{g_3}{28}, \quad a_3 = \frac{g_2^2}{1200}. \quad (5)$$

By integration of (2) one finds

$$\sigma'(u) = \left(\frac{1}{u} - \frac{a_1 u^3}{3} - \frac{a_2 u^5}{5} - \frac{a_3 u^7}{7} \dots \right) \sigma(u),$$

and from this equation follows

$$\alpha = 0, \quad \alpha_1 = -\frac{g_2}{240}, \quad \alpha_2 = -\frac{g_3}{840}, \quad \alpha_3 = -\frac{g_2^2}{161280}. \quad (6)$$

That $\alpha = 0$ follows also from the $\sigma'''(u) = 0$ proved in § 35; there, however, this was proved by a rather circuitous path. Thus we have the following beginning of the development of $\sigma(u)$, in which, for the sake of clarity, the numerical denominators are decomposed into their prime factors:

$$\sigma(u) = u - \frac{g_2}{2^4 \cdot 3 \cdot 5} u^5 - \frac{g_3}{2^3 \cdot 3 \cdot 5 \cdot 7} u^7 - \frac{g_2^2}{2^9 \cdot 3^2 \cdot 5 \cdot 7} u^9 - \dots \quad (7)$$

For the recurrent calculation of the coefficients in the power series for $\sigma(u)$ there serves a partial differential equation which may be regarded as a transformation of the partial differential equation for the ϑ -functions. We derive it in the following way.

The function $\sigma(u)$ was determined up to a factor independent of u by the two period equations § 35,(9)

$$\sigma(u + \omega_1) = -e^{\eta_1(2u + \omega_1)} \sigma(u), \quad \sigma(u + \omega_2) = -e^{\eta_2(2u + \omega_2)} \sigma(u), \quad \eta_1 \omega_2 - \eta_2 \omega_1 = \pi i. \quad (8)$$

By differentiating the last equation (8), however, one obtains

$$\begin{aligned} \frac{\partial \eta_1}{\partial \omega_1} \omega_2 - \frac{\partial \eta_2}{\partial \omega_1} \omega_1 - \eta_2 &= 0, \\ \frac{\partial \eta_1}{\partial \omega_2} \omega_2 - \frac{\partial \eta_2}{\partial \omega_2} \omega_1 + \eta_1 &= 0, \end{aligned}$$

hence

$$\omega_2 \left(\eta_1 \frac{\partial \eta_1}{\partial \omega_1} + \eta_2 \frac{\partial \eta_1}{\partial \omega_2} \right) = \omega_1 \left(\eta_1 \frac{\partial \eta_2}{\partial \omega_1} + \eta_2 \frac{\partial \eta_2}{\partial \omega_2} \right),$$

and we therefore introduce a quantity r , which we define thus:

$$\eta_1 \frac{\partial \eta_1}{\partial \omega_1} + \eta_2 \frac{\partial \eta_1}{\partial \omega_2} = -r \omega_1, \quad \eta_1 \frac{\partial \eta_2}{\partial \omega_1} + \eta_2 \frac{\partial \eta_2}{\partial \omega_2} = -r \omega_2. \quad (9)$$

Now by differentiating the period equations (8) one proves that the function

$$S(u) = \frac{\partial^2 \sigma}{\partial u^2} + 4\eta_1 \frac{\partial \sigma}{\partial \omega_1} + 4\eta_2 \frac{\partial \sigma}{\partial \omega_2} + 4ru^2 \sigma$$

satisfies the period equations (8), and hence is equal to $C\sigma(u)$, where C is independent of u . But the development of $S(u)$ according to powers of u begins with u^3 , whereas $\sigma(u)$ begins with u ; consequently $C = 0$.

By (7) the developments of $\partial \sigma / \partial \omega_1$ and $\partial \sigma / \partial \omega_2$ begin only with u^5 , and the first terms of $\partial^2 \sigma / \partial u^2$ and $4ru^2 \sigma$ are $-g_2 u^3 / 12$ and $4ru^3$, whence $4r = g_2 / 12$ follows, and accordingly the differential equation for σ assumes the form

$$\frac{\partial^2 \sigma}{\partial u^2} + 4\eta_1 \frac{\partial \sigma}{\partial \omega_1} + 4\eta_2 \frac{\partial \sigma}{\partial \omega_2} + \frac{g_2}{12} u^2 \sigma = 0, \quad (10)$$

and at the same time from (9) there result the relations

$$\eta_1 \frac{\partial \eta_1}{\partial \omega_1} + \eta_2 \frac{\partial \eta_1}{\partial \omega_2} = -\frac{g_2}{48} \omega_1, \quad \eta_1 \frac{\partial \eta_2}{\partial \omega_1} + \eta_2 \frac{\partial \eta_2}{\partial \omega_2} = -\frac{g_2}{48} \omega_2. \quad (11)$$

Now in the differential equation (10) the variables g_2, g_3 are to be introduced in place of the variables ω_1, ω_2 ; if therefore we put

$$\frac{\partial \sigma}{\partial \omega_1} = \frac{\partial \sigma}{\partial g_2} \frac{\partial g_2}{\partial \omega_1} + \frac{\partial \sigma}{\partial g_3} \frac{\partial g_3}{\partial \omega_1}, \quad \frac{\partial \sigma}{\partial \omega_2} = \frac{\partial \sigma}{\partial g_2} \frac{\partial g_2}{\partial \omega_2} + \frac{\partial \sigma}{\partial g_3} \frac{\partial g_3}{\partial \omega_2},$$

then (10) assumes the form

$$\frac{\partial^2 \sigma}{\partial u^2} + h_2 \frac{\partial \sigma}{\partial g_2} + h_3 \frac{\partial \sigma}{\partial g_3} + \frac{g_2}{12} u^2 \sigma = 0, \quad (12)$$

where

$$h_2 = 4 \left(\eta_1 \frac{\partial g_2}{\partial \omega_1} + \eta_2 \frac{\partial g_2}{\partial \omega_2} \right), \quad h_3 = 4 \left(\eta_1 \frac{\partial g_3}{\partial \omega_1} + \eta_2 \frac{\partial g_3}{\partial \omega_2} \right).$$

The quantities h_2, h_3 , expressed by g_2, g_3 , now result from the differential equation itself if one inserts for σ the development (7). According to this, up to the seventh power of u ,

$$\begin{aligned} \frac{\partial^2 \sigma}{\partial u^2} &= -\frac{g_2}{12} u^3 - 12g_3 \frac{u^5}{2^4 \cdot 3 \cdot 5} - \frac{3}{8} g_2^2 \frac{u^7}{2^3 \cdot 3 \cdot 5 \cdot 7}, \\ \frac{\partial \sigma}{\partial g_2} &= -\frac{u^5}{2^4 \cdot 3 \cdot 5}, \quad \frac{\partial \sigma}{\partial g_3} = -\frac{u^7}{2^3 \cdot 3 \cdot 5 \cdot 7}, \\ \frac{g_2}{12} u^2 \sigma &= \frac{g_2}{12} u^3 - \frac{7}{24} g_2^2 \frac{u^7}{2^3 \cdot 3 \cdot 5 \cdot 7}, \end{aligned}$$

and if this is inserted in (12), and the coefficients of u^5 and u^7 are set equal to zero, there results

$$h_2 = -12g_3, \quad h_3 = -\frac{2}{3} g_2^2,$$

and we obtain the differential equation

$$\frac{\partial^2 \sigma}{\partial u^2} - 12g_3 \frac{\partial \sigma}{\partial g_2} - \frac{2}{3} g_2^2 \frac{\partial \sigma}{\partial g_3} + \frac{g_2}{12} u^2 \sigma = 0. \quad (13)$$

If one writes the series development for σ with undetermined coefficients as

$$\sigma = \sum A_\nu \frac{u^{2\nu+1}}{(2\nu+1)!}, \quad (14)$$

then from (13) one obtains the recursion formula

$$A_\nu = 12g_3 \frac{\partial A_{\nu-1}}{\partial g_2} + \frac{2}{3} g_2^2 \frac{\partial A_{\nu-1}}{\partial g_3} - \frac{(2\nu-1)(2\nu-2)}{12} g_2 A_{\nu-2}, \quad (15)$$

from which it is to be concluded that the A_ν are integral rational functions of g_2 and g_3 , whose coefficients are rational numbers that can have only powers of 2 and of 3 in the denominator. The series (14) therefore consists of terms of the form

$$a_{m,n} \left(\frac{1}{2} g_2 \right)^m (2g_3)^n \frac{u^{2\nu+1}}{(2\nu+1)!},$$

and from the homogeneity of the function σ [§ 37,(8), § 46,(13)] it follows that

$$2m + 3n = \nu. \quad (16)$$

For the coefficients $a_{m,n}$ one then obtains from (15):

$$\begin{aligned} a_{m,n} &= 3(m+1)a_{m+1,n-1} + \frac{16}{3}(n+1)a_{m-2,n+1} \\ &\quad - \frac{1}{3}(2m+3n-1)(4m+6n-1)a_{m-1,n}, \end{aligned} \quad (17)$$

where $a_{0,0} = 1$, and those a for which one of the two indices is negative are to be put equal to zero.

From this one can find $a_{m,n}$ if the earlier $a_{m,n}$, that is, those for which $2m + 3n$ is smaller than ν , have already been calculated. It is seen from this that the $a_{m,n}$ contain only powers of 3 in the denominator. In the *Formeln und Lehrsätzen zum Gebrauch der elliptischen Funktionen*, edited by H. A. Schwarz, after lectures by Weierstrass (2nd edition, 1893), the $a_{m,n}$ have been calculated up to $\nu = 17$. It appears there that these coefficients are not only whole numbers, but also contain, as ν increases, increasing powers of 3 as factors. I am not able to give a proof of this.